

CS330 Operating Systems and Lab.

Mass-Storage Systems

Spring 2026

KAIST

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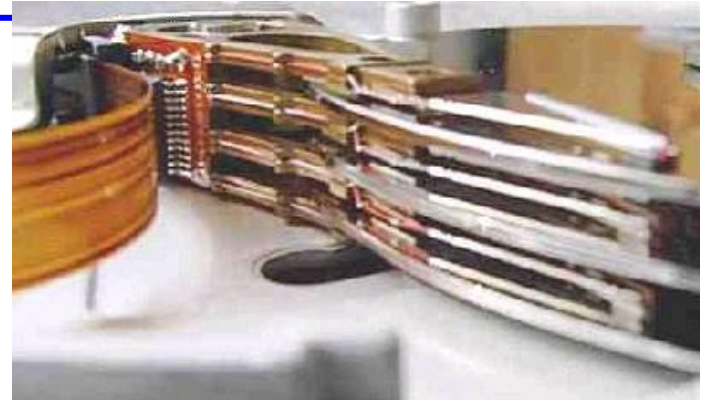
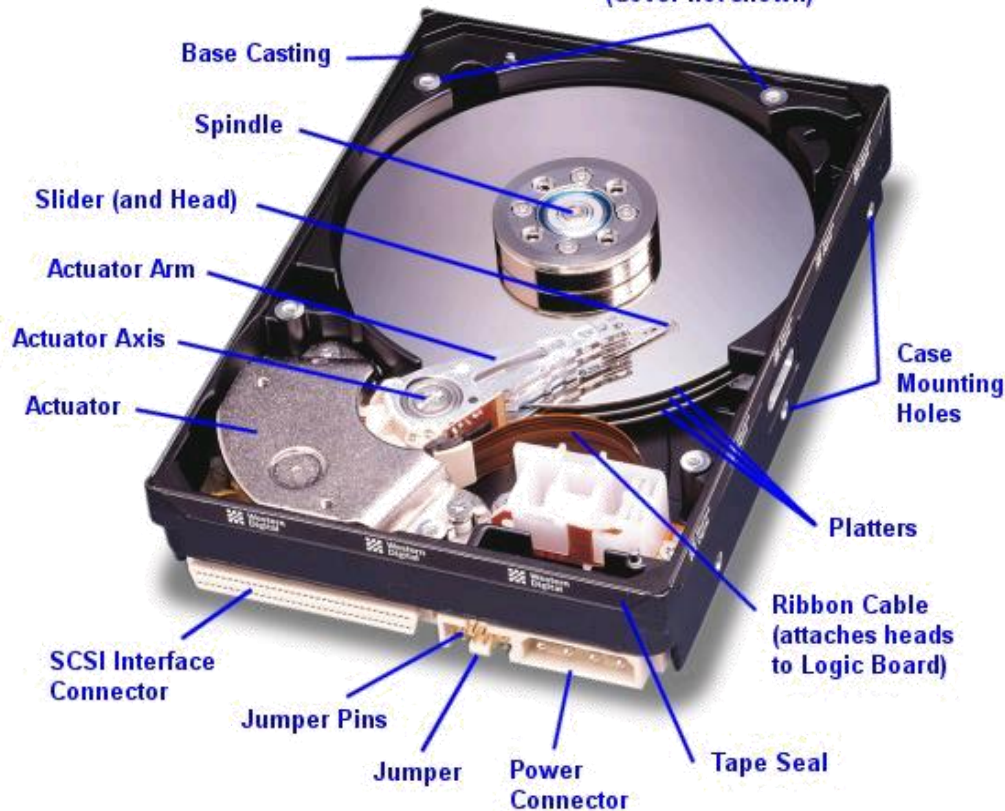
Mass-Storage Systems

- Disks and SSDs



Hard Disk Drives (HDDs)

Cover Mounting Holes
(Cover not shown)



**Read/Write Head
Side View**



IBM/Hitachi Microdrive

IBM Personal Computer/AT (1986)
30 MB hard disk - \$500
30-40ms seek time
0.7-1 MB/s (est.)

What's inside of a hard drive?



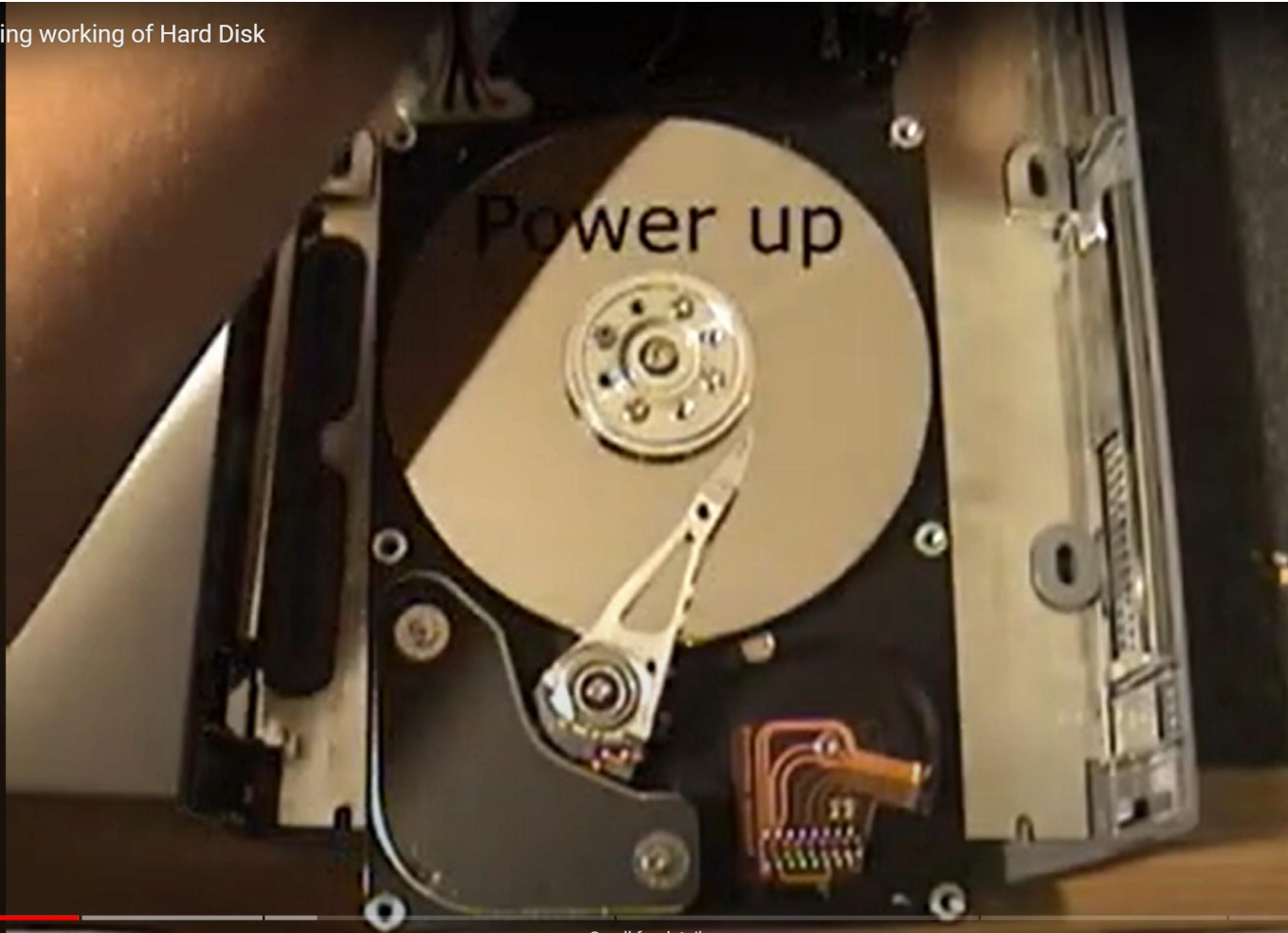
Hard drives are sealed so that no dust can get inside, once you take yours apart it most

▶ ▶ 🔊 0:11 / 1:43

Scroll for details
▼



The amazing working of Hard Disk



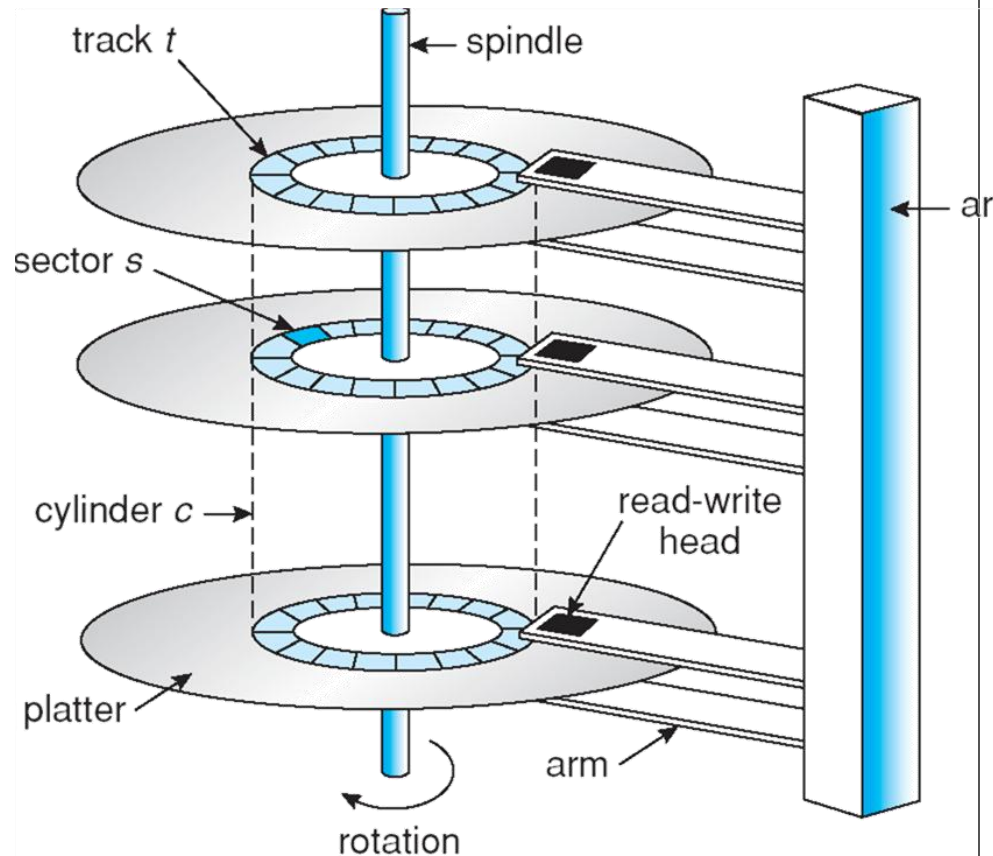
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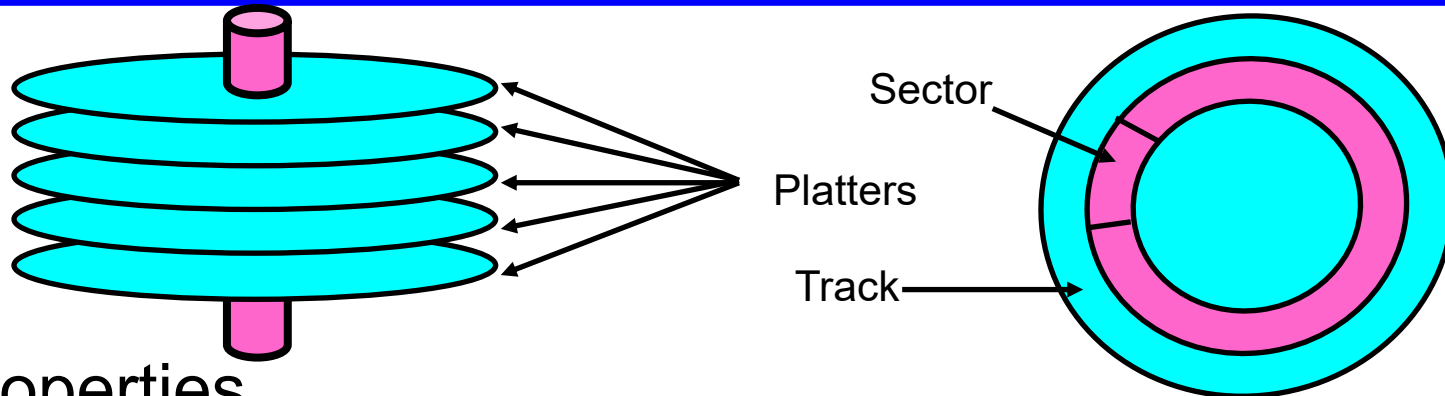


Disk

- Stack of magnetic platters
 - Rotate together on a central spindle at 3,600-15,000 RPM
- Disk arm assembly
 - Arms rotate around pivot, all move together
 - Arms contain disk heads – one for each recording surface
 - Heads read and write data to platters

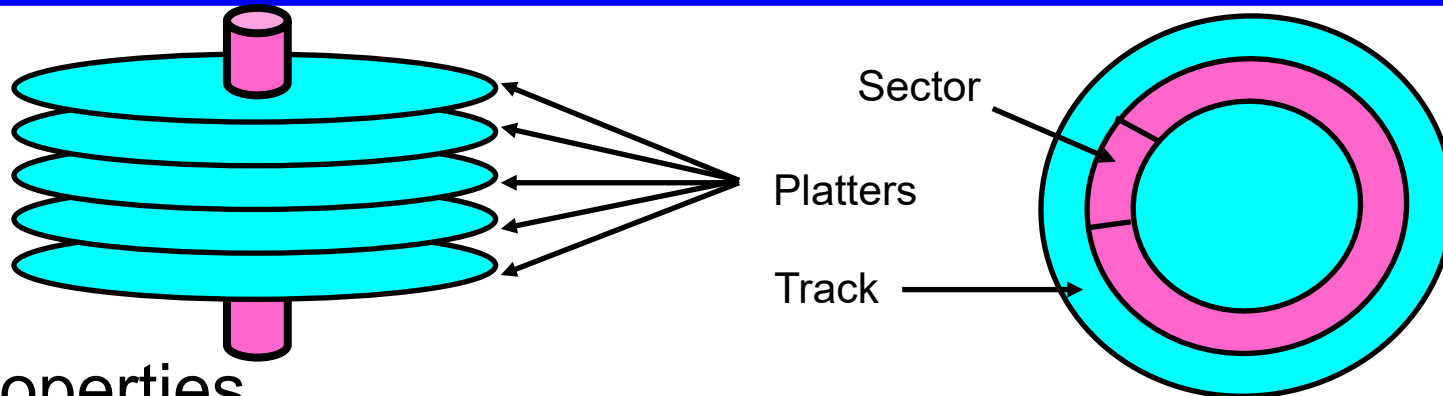


Properties of a Magnetic Hard Disk



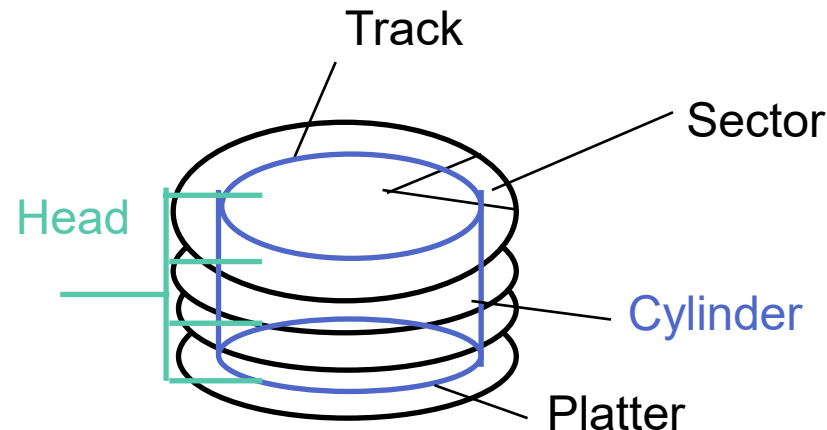
- Properties
 - Independently addressable element: **sector**
 - OS always transfers groups of sectors together—“**blocks**”
 - A disk can access directly any given block either sequentially or randomly.
- Typical numbers (depending on the disk size):
 - 500 to more than 20,000 tracks per surface
 - 32 to 800 sectors per track
- Zoned bit recording
 - Constant bit density: more bits (sectors) on outer tracks

Properties of a Magnetic Hard Disk



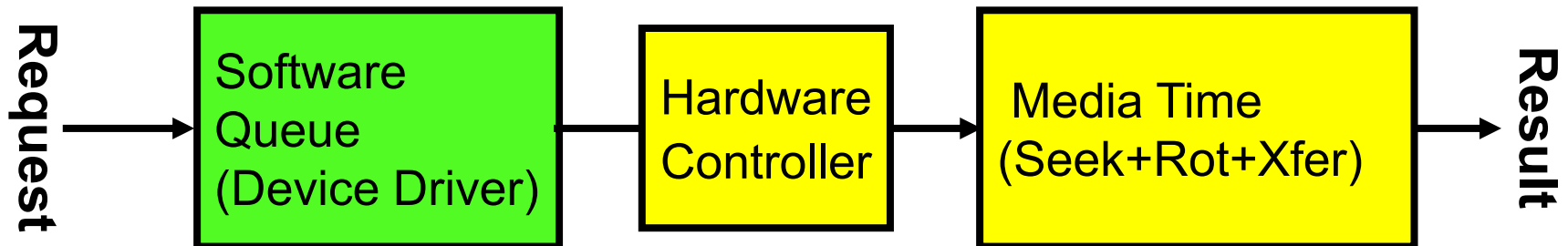
- Properties

- Independently addressable element: **sector**
 - OS always transfers groups of sectors together—“**blocks**”
- Cylinder: all the tracks under the head at a given point on all surfaces



Magnetic Disk Characteristic

- Read/write: three-stage process:
 - **Seek time**: position the head/arm over the proper track (into proper cylinder)
 - **Rotational latency**: wait for the desired sector to rotate under the read/write head
 - **Transfer time**: transfer a block of bits (sector) under the read-write head
- **Disk Latency = Queuing Time + Controller time + Seek Time + Rotation Time + Xfer Time**



- **Highest Bandwidth:**
 - Transfer large group of blocks sequentially from one track

Typical Numbers of a Magnetic Disk

Parameter	Info / Range
Average seek time	Typically 5-10 milliseconds. Depending on reference locality, actual cost may be 25-33% of this number.
Average rotational latency	Most laptop/desktop disks rotate at 3600-7200 RPM (8-16 ms/rotation). Server disks up to 15,000 RPM. Average latency is halfway around disk yielding corresponding times of 4-8 milliseconds
Controller time	Depends on controller hardware
Transfer time	Typically 50 to 100 MB/s. Depends on: • Transfer size (usually a sector): 512B – 1KB per sector • Rotation speed: 3600 RPM to 15000 RPM • Recording density: bits per inch on a track • Diameter: ranges from 1 in to 5.25 in
Cost	Drops by a factor of two every 1.5 years (or even faster). \$0.02/GB

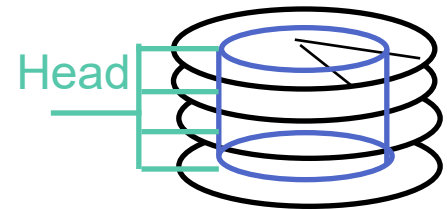
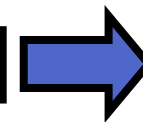
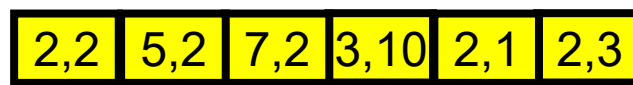
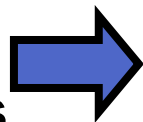
Disk Performance Examples

- Assumptions:
 - Ignoring queuing and controller times for now
 - Avg seek time of 5ms,
 - 7200RPM $\Rightarrow$ Time for one rotation: $60000\text{ms}/7200 \approx 8\text{ms}$
 - Transfer rate of 4MByte/s, sector size of 1 KByte
- Read sector from random place on disk:
 - Seek (5ms) + Rot. Delay (4ms) + Transfer (0.25ms)
 - Approx 10ms to fetch/put data: **100 KByte/sec**
- Read sector from random place in same cylinder:
 - Rot. Delay (4ms) + Transfer (0.25ms)
 - Approx 5ms to fetch/put data: **200 KByte/sec**
- Read next sector on same track:
 - Transfer (0.25ms): **4 MByte/sec**
- Key to using disk effectively (especially for file systems) is to *minimize seek and rotational delays*

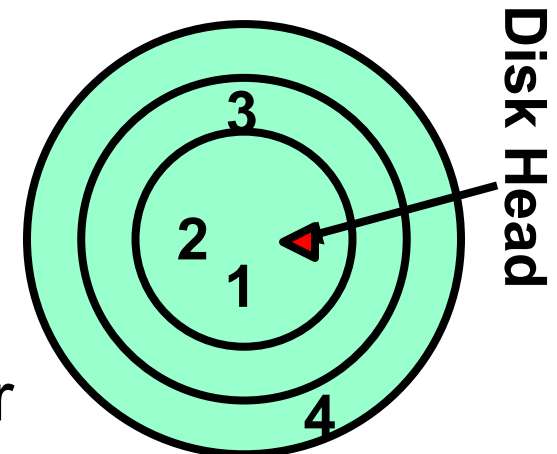
Disk Scheduling

- Disk can do only one request at a time; What order do you choose when handling queued requests?
 - Request denoted by (track, sector)

User
Requests

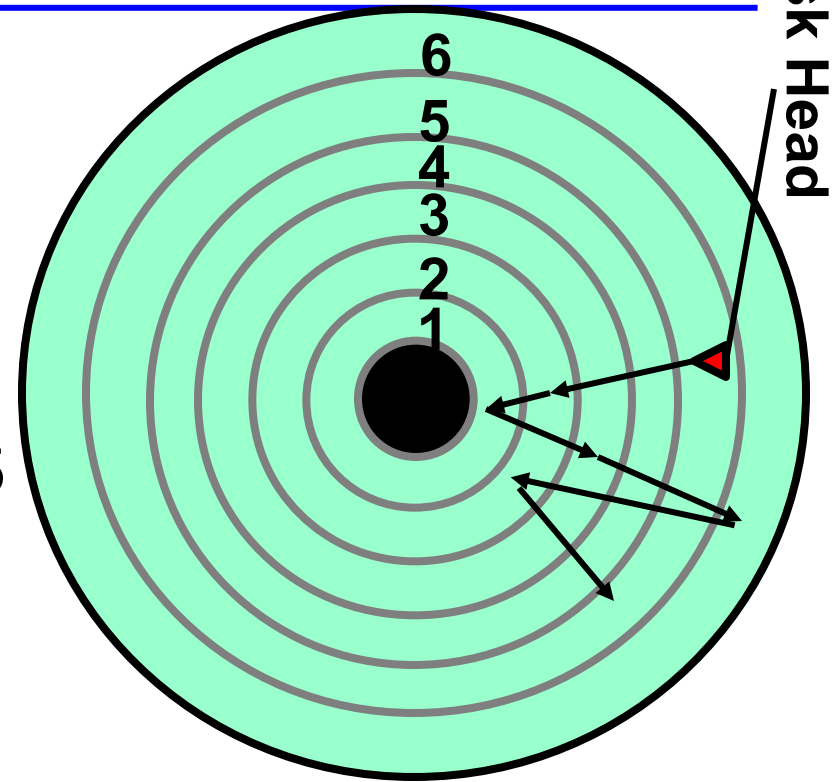


- Scheduling algorithms:
 - First In First Out (FIFO)
 - Shortest Seek Time First
 - SCAN
 - C-SCAN
- In our examples we will ignore the sector
 - Consider only track #



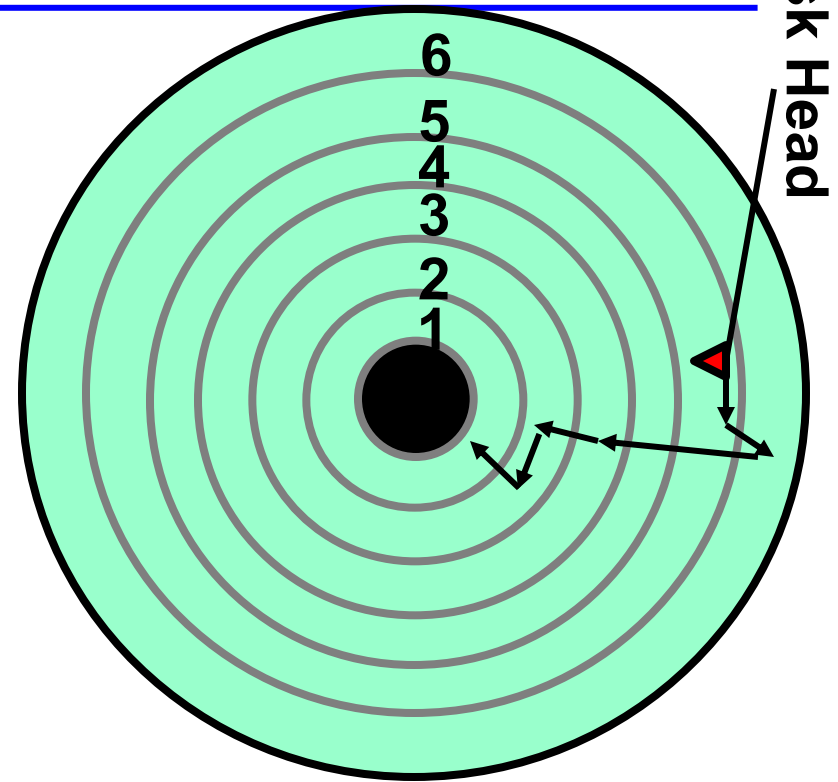
FIFO: First In First Out

- Schedule request in the order they arrive in the queue
- Example:
 - Request queue: 2, 1, 3, 6, 2, 5
 - Scheduling order: 2, 1, 3, 6, 2, 5
- Pros: Fair among requesters
- Cons: Order of arrival may be to random spots on the disk $\Rightarrow$ Very long seeks



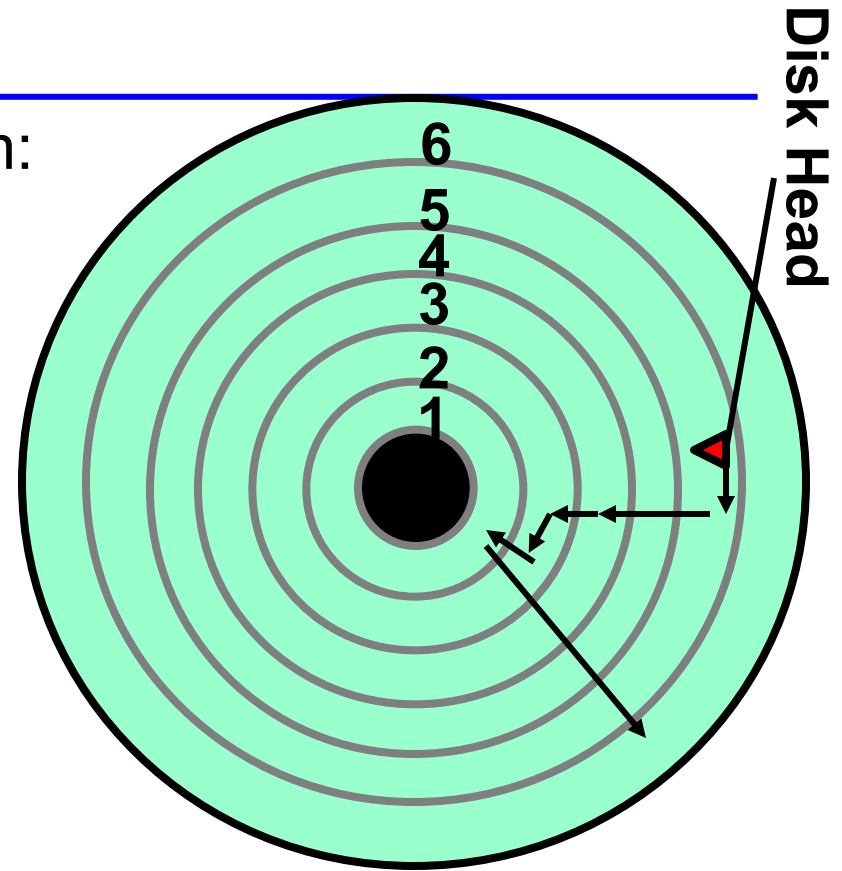
SSTF: Shortest Seek Time First

- Pick the request that's closest to the head on the disk
 - Although called SSTF, include rotational delay in calculation, as rotation can be as long as seek
- Example:
 - Request queue: 2, 1, 3, 6, 2, 5
 - Scheduling order: 5, 6, 3, 2, 2, 1
- Pros: reduce seeks
- Cons: may lead to starvation



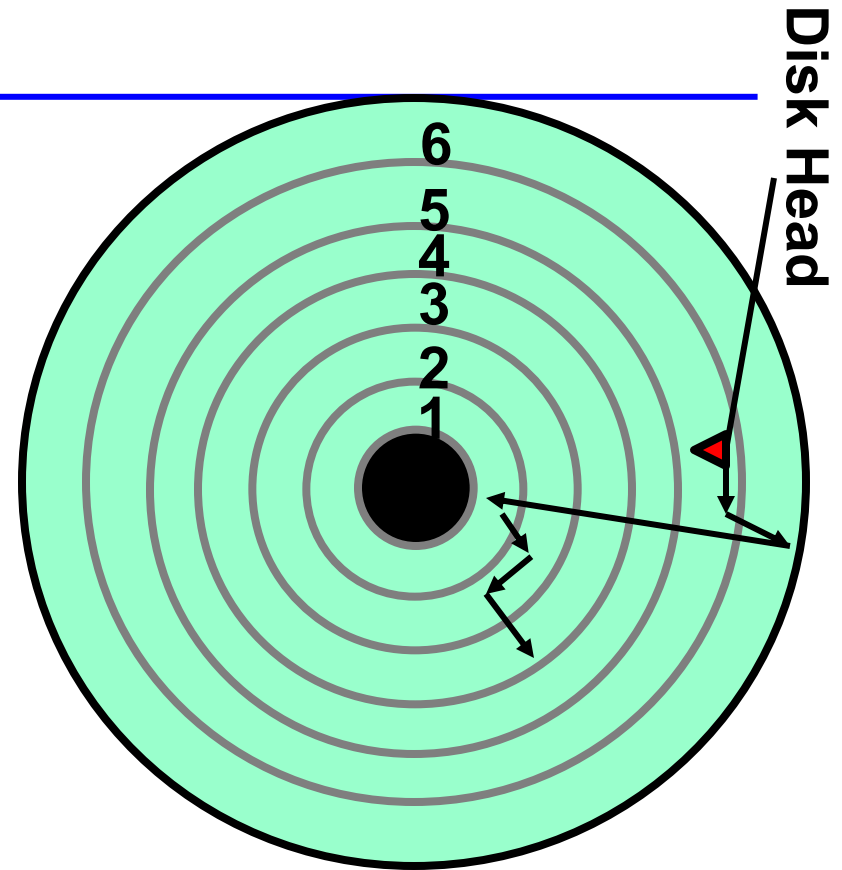
SCAN

- Implements an Elevator Algorithm: take the closest request in the direction of travel
- Example:
 - Request queue: 2, 1, 3, 6, 2, 5
 - Head is moving towards center
 - Scheduling order: 5, 3, 2, 2, 1, 6
- Pros:
 - No starvation
 - Low seek
- Cons: favor middle tracks

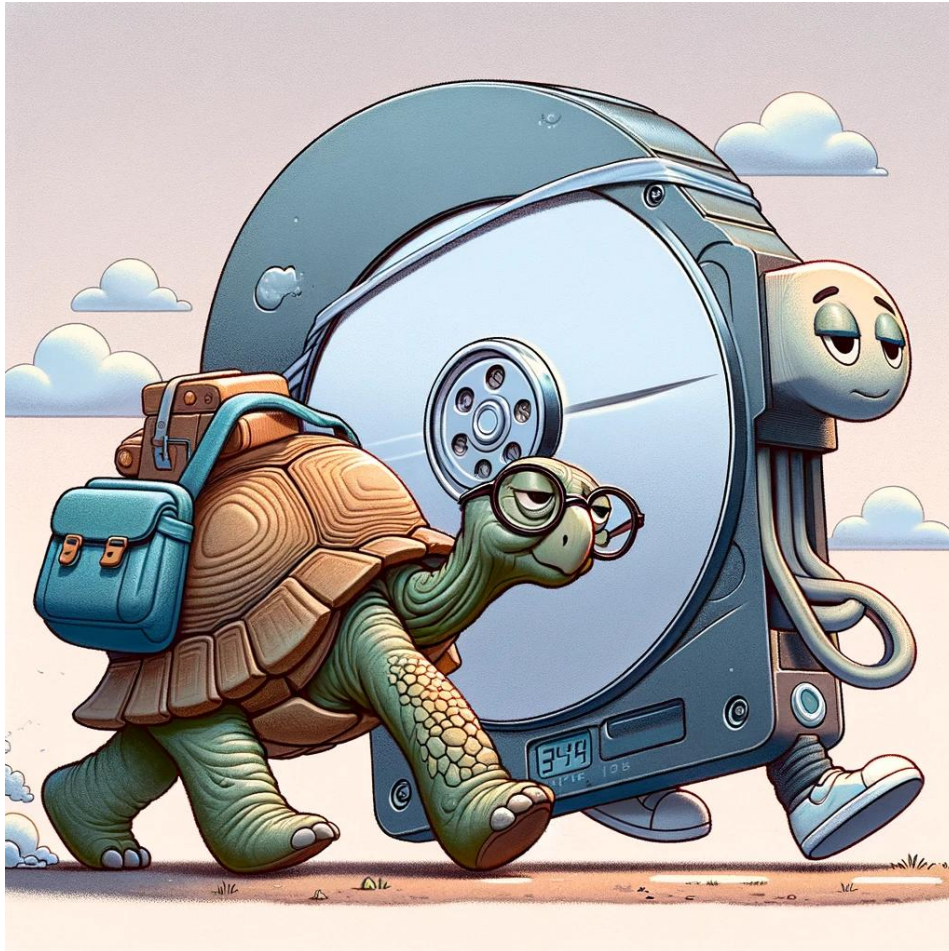


C-SCAN

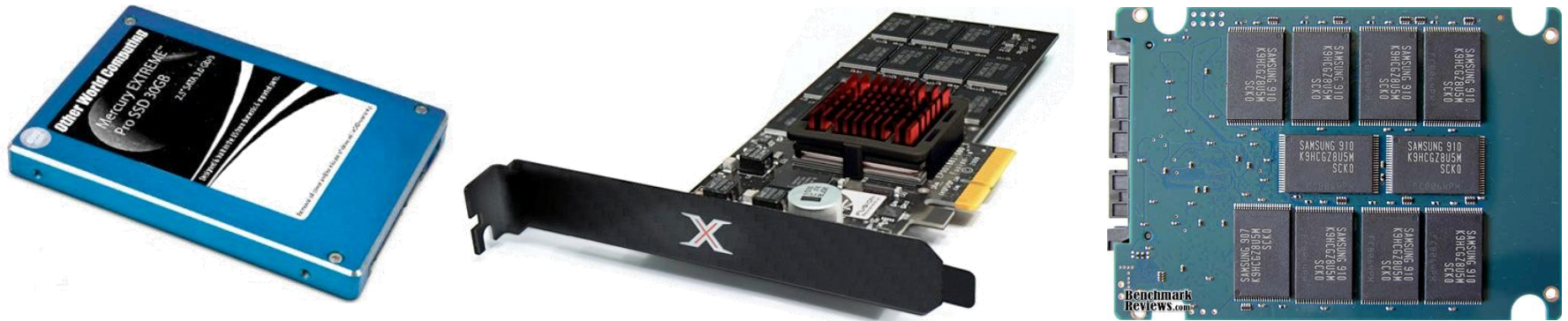
- Like SCAN but only serves request in only one direction
- Example:
 - Request queue: 2, 1, 3, 6, 2, 5
 - Head only serves request on its way from center towards edge
 - Scheduling order: 5, 6, 1, 2, 2, 3
- Pros:
 - Fairer than SCAN
- Cons: longer seeks on the way back



Hard Disks are Slow!

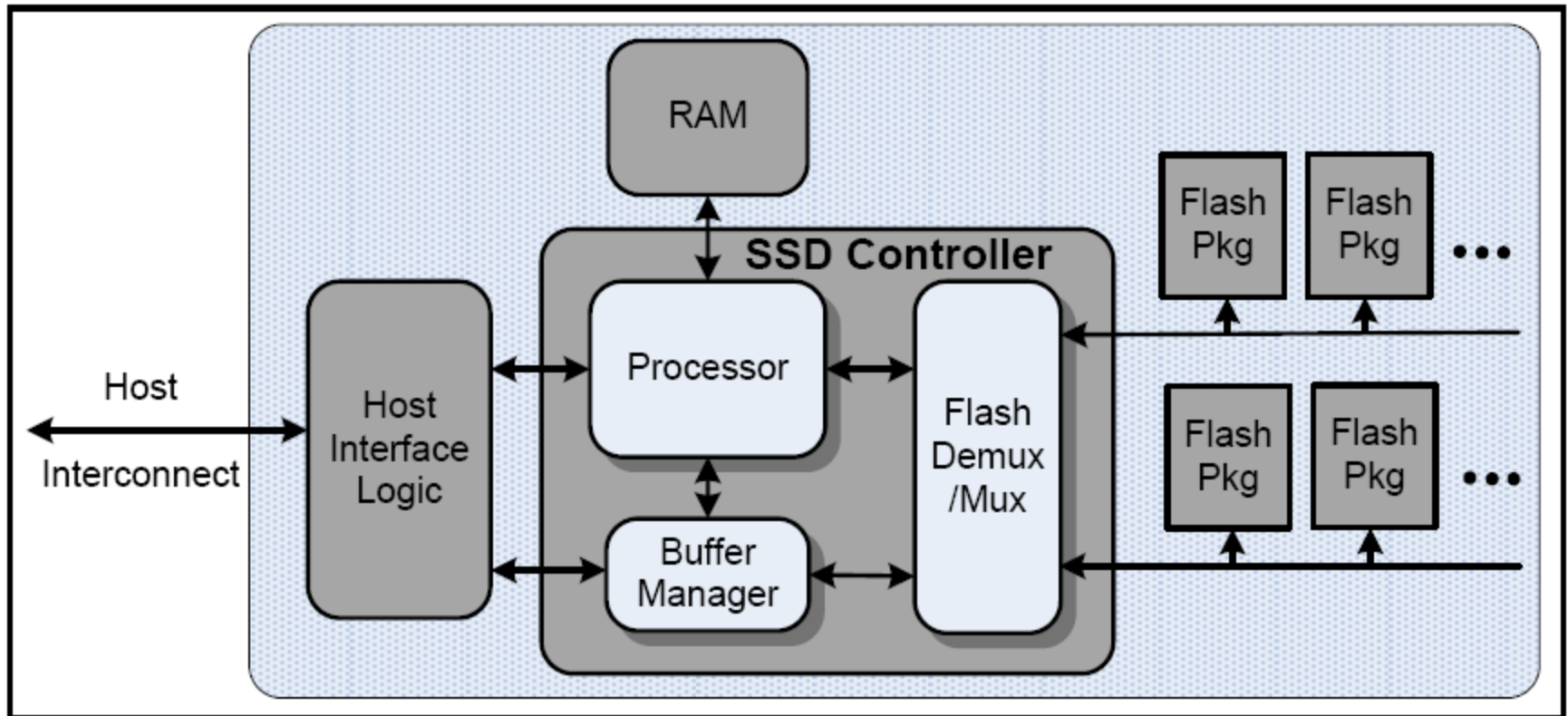


Solid State Disks (SSDs)

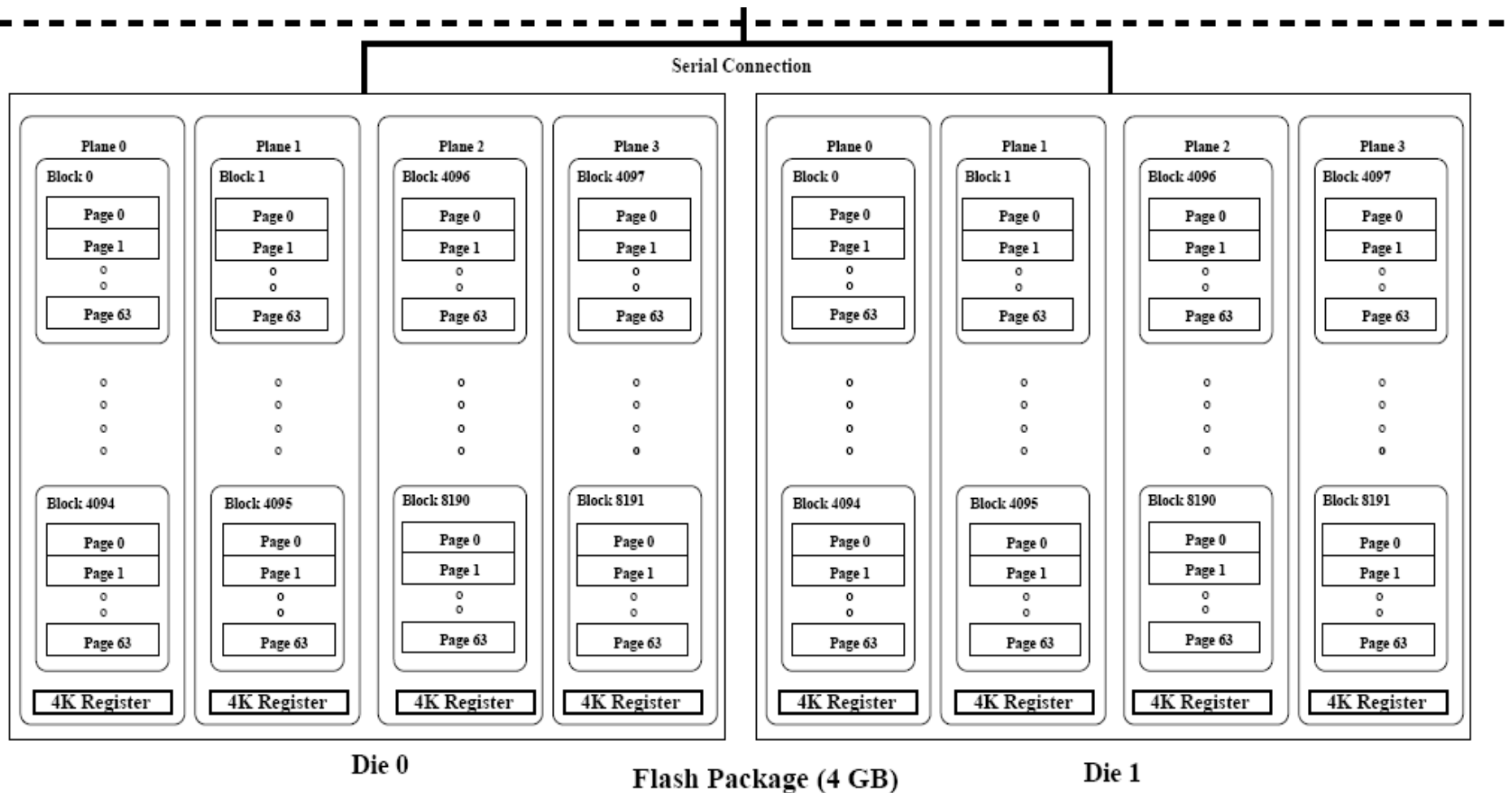


- 1995 – Replace rotating magnetic media with non-volatile memory (battery backed DRAM)
 - Since 2009, use NAND Flash: Single Level Cell (1-bit/cell), Multi-Level Cell (2-bit/cell)
- Sector addressable, but stores 4-64 “sectors” per memory page
- No moving parts (no rotate/seek motors)
 - Eliminates seek and rotational delay (0.1-0.2ms access time)
 - Very low power and lightweight

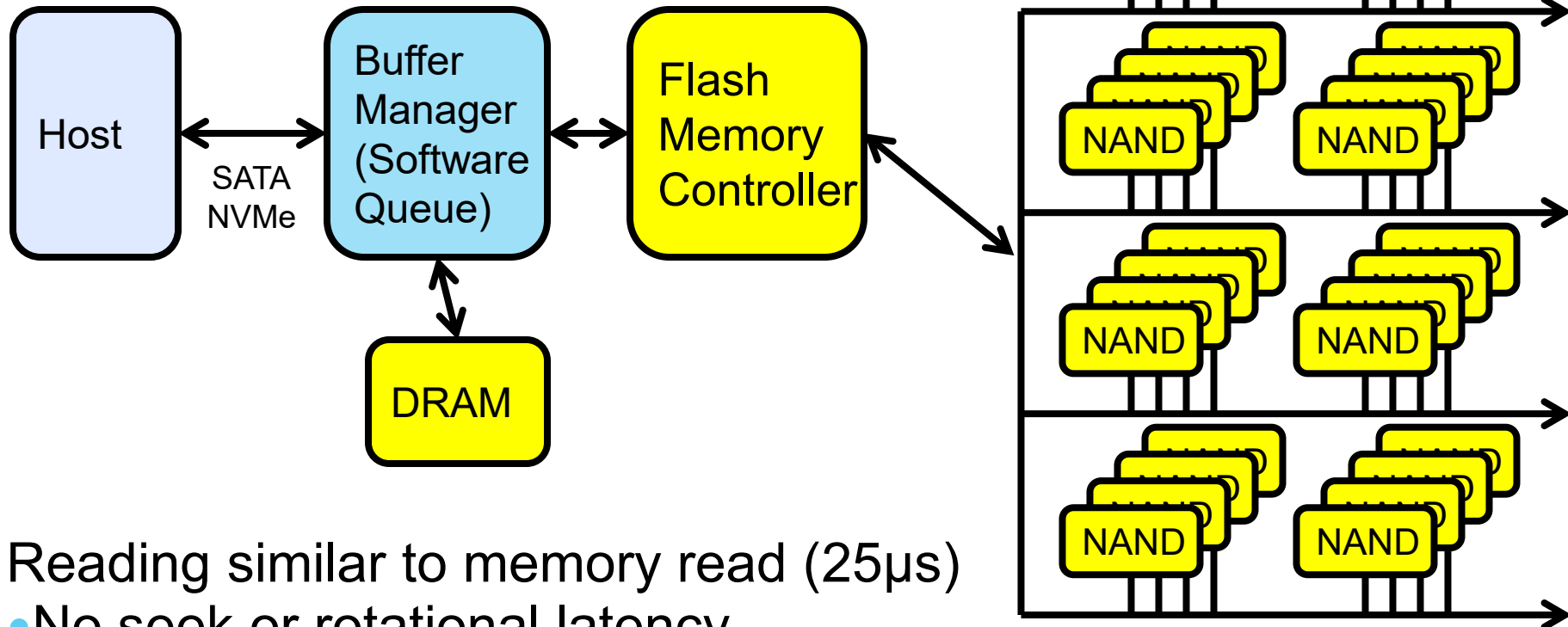
SSD Logic components



Flash Internals



SSD Architecture – Reads



Reading similar to memory read (25μs)

- No seek or rotational latency
- Transfer time: transfer a block of bits (sector)
 - Limited by controller and disk interface (SATA: 300-600MB/s, NVMe: 3,500MB/s)
- **Latency = Queuing Time + Controller time + Xfer Time**
- **Highest Bandwidth:** Sequential OR Random reads

SSD Architecture – Writes

- Writing data is complex! ($\sim 200\mu\text{s}$ – 1.7ms)
- Can only write empty pages (erase takes $\sim 1.5\text{ms}$)
- Controller maintains pool of empty pages by coalescing used sectors (read, erase, write), also reserve some % of capacity
- Write and erase cycles require “high” voltage
- Damages memory cells, limits SSD lifespan
- Controller uses ECC, performs wear leveling
- Result is very workload dependent performance
- **Latency = Queuing Time + Controller time (Find Free Block) + Xfer Time**
- **Highest BW:** Seq. OR Random writes (limited by empty pages)
 - Sequential easier to implement since can write all data to same pg

Rule of thumb: writes 10x more expensive than reads, and erases 10x more expensive than writes

Storage Performance & Price

	Bandwidth (sequential R/W)	Cost/GB	Size
HDD	50-100 MB/s	\$0.01-0.05/GB	2-10 TB
SSD	200-600 MB/s (SATA) 6 GB/s (PCI)	\$0.05-0.1/GB	512GB-4TB
DRAM	10-16 GB/s	\$5/GB	4GB-64GB

BW: SSD up to x10 than HDD, DRAM > x10 than SSD
Price: HDD x5 less than SSD, SSD x10 less than DRAM

Quiz 12.3: HDDs and SSDs

- Q1: True _ False _ The block is the smallest addressable unit on a disk
- Q2: True _ False _ An SSD has zero seek time
- Q3: True _ False _ For an HDD, the read and write latencies are similar
- Q4: True _ False _ For an SSD, the read and write latencies are similar
- Q5: Consider the following sequence of requests (2, 4, 1, 8), and assume the head position is on track 9. Then, the order in which SSTF services the requests is

SSD Summary

- Pros (vs. hard disk drives):
 - Low latency, high throughput (eliminate seek/rotational delay)
 - No moving parts:
 - Very light weight, low power, silent, very shock insensitive
 - Read at memory speeds (limited by controller and I/O bus)
- Cons
 - Expensive (3-10x disk)
 - Hybrid alternative: combine small SSD with large HDD
 - Asymmetric block write performance: read pg/erase/write pg
 - Controller garbage collection (GC) algorithms have major effect on performance
 - Limited drive lifetime
 - 1-10K writes/page for MLC NAND
 - Avg failure rate is 6 years, life expectancy is 9–11 years
- These are changing rapidly!